
NYCU IMF - 114 Financial Econometrics

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Question 18

Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- (a) Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?
- The percentage of **mothers** have some college education is 12.14953%,
 - and the percentage of **fathers** have some college education is 11.68224%;
 - percentage of **both parents** have some college education is 5.140187%,
 - percentage of **at least one parent** have some college education is 18.69159%,
- (b) Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

1. Interpreting the Magnitudes

- $Corr(EDUC, MOTHERCOLL) \approx 0.3594705$
- $Corr(EDUC, FATHERCOLL) \approx 0.3984962$

Are the magnitudes important?

Yes, they are critically important. The magnitude directly measures the instrument relevance condition. For an instrumental variable to be useful, it must have a strong, non-zero correlation with the endogenous explanatory variable (*EDUC*).

In cross-sectional microeconomic data (like the *mroz* dataset), correlations of 0.36 and 0.40 are considered quite solid. They are positive and meaningfully far from zero. This indicates that having a mother or father who attended college is strongly associated with the woman herself acquiring more years of education.

Because these magnitudes are reasonably large, this gives us initial confidence that *MOTHERCOLL* and *FATHERCOLL* are not weak instruments. (Weak instruments are dangerous because they cause 2SLS estimates to become biased towards OLS and severely inflate standard errors).

2. The Logical Argument

- The Threshold (Credential) Effect: A linear, continuous variable assumes that the jump from 8th to 9th grade has the exact same impact on a household's educational culture as the jump from 12th grade to freshman year of college. Logically, this is flawed. Entering college represents a major structural shift in a family's socioeconomic status, expectations, and resources. A dummy variable captures this discrete "college threshold" effect much more cleanly than a continuous count of years.
 - Reducing Measurement Error: Survey respondents often misremember or guess the exact number of years their parents went to school (e.g., confusing 10 years with 11 years). This introduces "classical measurement error," which attenuates correlations. However, whether a parent went to college or not is a highly salient family fact that is rarely misreported. By converting a noisy continuous variable into a sharp binary indicator, we reduce measurement error and potentially isolate a stronger signal for the first-stage regression.
- (c) Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

The wage equation in Example 10.5:

$$\ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + e$$

The 95% Interval Estimate

For the model using *MOTHERCOLL* as instrumental variables, the 95% confidence interval for the coefficient of *EDUC* is:

$$[-0.001220, 0.1533]$$

- (d) For the problem in part (c), estimate the first-stage equation. What is the value of the F-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?

First stage equation:

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHERCOLL + v$$

1. The F-test Statistic

The value of the F-test statistic for the hypothesis that *MOTHERCOLL* = 0 (has no

effect on *EDUC*) is 63.563.

2. Is *MOTHERCOLL* a strong instrument?

Yes, it is a very strong instrument.

The standard econometric rule of thumb dictates that an F-statistic on the excluded instrument(s) must be greater than 10 to avoid the weak instrument problem. Since $63.563 \gg 10$, we easily reject the null hypothesis. We can confidently conclude that a mother's college attendance has a highly significant explanatory power over the daughter's years of education. Additionally, the coefficient itself makes intuitive economic sense: the Estimate of 2.517 indicates that having a mother who attended college is associated with roughly 2.5 additional years of education for the woman in the sample.

- (e) Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

1. The 95% Interval Estimate

For the model using both *MOTHERCOLL* and *FATHERCOLL* as instrumental variables, the 95% confidence interval for the coefficient of *EDUC* is:

$$[0.02751845, 0.1481769]$$

2. Is it narrower or wider than the one in part (c)?

It is narrower.

We can verify this by looking at the width of the two intervals:

- Part (c) Width (Single IV): $0.1533 - (-0.0012) = 0.1545$
- Part (e) Width (Two IVs): $0.1482 - 0.0275 = 0.1207$

Why did this happen? (The Econometric Intuition)

Adding a second valid instrument increases the amount of exogenous variation you are using to isolate the "clean" part of *EDUC*. In econometrics, this increases the efficiency of your Two-Stage Least Squares (2SLS) estimator. Greater efficiency means smaller standard errors. Because the confidence interval is calculated using the standard error (Estimate $\pm 1.96 \times$ Standard Error), a smaller standard error mathematically shrinks the interval, making it narrower and your estimate more precise. Notice a crucial outcome here: The new interval no longer contains zero. By adding the second instrument and gaining efficiency, our estimate for the return to education is now statistically significant at the 5% level.

- (f) For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?

First stage equation:

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHERCOLL + \theta_2 FATHERCOLL + v$$

1. Testing Joint Significance

The F-statistic for the joint hypothesis that $MOTHERCOLL = 0$ and $FATHERCOLL = 0$ is 56.963. The p-value is extremely close to zero ($< 2.2e - 16$), meaning we overwhelmingly reject the null hypothesis that these two variables have no combined effect on education.

2. Do these instruments seem adequately strong?

Yes, they are very strong. Once again, we apply the standard econometric rule of thumb: the F-statistic on the excluded instruments in the first-stage regression should be greater than 10. Since 56.963 is much greater than 10, we can confidently state that a mother's and father's college attendance, taken together, provide a very strong source of exogenous variation to predict a daughter's education. We do not have a weak instrument problem here.

- (g) For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

1. Identifying the Test

The test for the validity of the surplus instrument in an overidentified model is the Sargan test (also known as the Overidentifying Restrictions Test or Hansen's J-test).

- Sargan statistic: 0.238
- p-value: 0.626

2. The Hypotheses

- Null Hypothesis (H_0): All instruments are valid. This means the instruments ($MOTHERCOLL$ and $FATHERCOLL$) are uncorrelated with the structural error term in the wage equation.
- Alternative Hypothesis (H_1): At least one instrument is invalid (correlated with the error term).

3. Conclusion

Because the p-value (0.626) is significantly greater than the standard alpha level of 0.05, we fail to reject the null hypothesis.

We can conclude that the instruments are valid. There is no statistical evidence to suggest that either a mother's or father's college attendance is correlated with the unobserved factors (the error term) affecting a woman's wage, other than through their impact on her education. The surplus instrument is valid, and the overidentified 2SLS model from part (e) is econometrically sound.

Question 20

The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- (a) Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

Based on the OLS output, the coefficient for excess return of the market portfolio (which is the estimated β) is 1.201840. Because $\beta > 1$, Microsoft stock over this period is classified as risky relative to the market portfolio. Economically, this means Microsoft's returns were more volatile than the broader market—specifically, for every 1% increase (or decrease) in the market's excess return, Microsoft's excess return tended to move in the same direction by approximately 1.20%. This is highly typical for high-growth technology equities, which tend to amplify broader market movements.

- (b) It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create *RANK* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the R^2 of the first-stage regression? Can *RANK* be regarded as a strong IV?

Conditions IV1–IV3:

- IV1 (Relevance): Does it correlate with the endogenous variable? Yes, a variable's rank is mechanically and strongly correlated with the variable itself.
- IV2 (Exogeneity): Is it uncorrelated with the error term? If the measurement error is truly random noise, it might change the precise decimal values of the returns but is unlikely to drastically scramble the overall sorted order (the rank) of 180 months of data.
- IV3 (Exclusion): The arbitrary integer rank (1 to 180) has no direct structural effect on Microsoft's returns; only the actual financial market return does.

1. Is the coefficient of *RANK* very significant?

Yes, it is extremely significant.

The t-value for the RANK coefficient is 43.1, and the corresponding p-value is $< 2e-16$ (essentially zero). This means the relationship between the market return and its ordinal rank is highly statistically significant.

2. What is the R^2 of the first-stage regression?

The Multiple R^2 is 0.9126.

This means that simply knowing the ordinal rank (1 to 180) explains over 91% of the variation in the actual market excess return.

3. Can RANK be regarded as a strong IV?

Absolutely. It is an exceptionally strong instrument.

The F-statistic on the instrument is 1858. Recall the econometric rule of thumb: an F-statistic greater than 10 indicates a strong instrument. A value of 1858 overwhelmingly rejects any concern of a weak instrument problem. Because RANK is essentially a monotonically transformed version of the endogenous variable itself, it mathematically guarantees a massive first-stage relationship.

- (c) Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

If the market return (mkt_{ex}) is truly measured with error, it will be correlated with the structural error term in the main CAPM equation, making it endogenous.

To test this, we take the unpredicted "noise" from the first stage—the residuals ($\hat{v}$)—and plug them directly into the original CAPM equation.

$$msft_{ex} = \alpha + \beta_1(mkt_{ex}) + \beta_2(\hat{v}) + error$$

- If the coefficient on $\hat{v}$ is statistically significant, it proves that the endogenous variable is carrying "dirty" variation correlated with the error term. Therefore, the variable is endogenous.
- If $\hat{v}$ is not significant, there is no evidence of endogeneity, and OLS is perfectly fine.

The Hausman Test for Endogeneity

The augmented regression result:

- The p-value: 0.0428.
- The 1% Significance Level: $\alpha = 0.01$
- Conclusion: Because $0.0428 > 0.01$, we fail to reject the null hypothesis.

Can we conclude that the market return is exogenous?

Yes, strictly at the 1% level of significance, we conclude that the market return is exogenous. There is not enough statistical evidence at this strict threshold to prove

that *mkt_ex* is correlated with the error term.

- (d) Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

1. The IV/2SLS Estimate

Looking at the output, the new estimated coefficient for *mkt_ex* using *RANK* as an instrumental variable is 1.278318.

2. Comparing IV to OLS

- OLS Estimate (Part a): 1.2018
- IV Estimate (Part d): 1.2783

The IV estimate is noticeably larger than the OLS estimate.

3. Does this agree with expectations?

Yes, it perfectly agrees with theoretical expectations.

As noted in the problem description, we suspected the market return was measured with error. In econometrics, "Classical Measurement Error" in an explanatory variable causes attenuation bias, which means the OLS estimator is biased toward zero.

Because the true relationship between Microsoft and the market is positive, the measurement error artificially dragged the OLS estimate down toward zero (yielding 1.2018). By using the instrument *RANK* to isolate the variation that is free from measurement error, the IV estimator corrected this attenuation bias, pushing the estimate back up to its "truer," higher value of 1.2783.

- (e) Create a new variable $POS = 1$ if the market return ($r_m - r_f$) is positive, and zero otherwise. Obtain the first-stage regression results using both *RANK* and *POS* as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

1. Testing Joint Significance

The F-statistic for the joint hypothesis that $RANK = 0$ and $POS = 0$ is 951.26, with a p-value of $< 2.2e - 16$. We overwhelmingly reject the null hypothesis that these instruments have no combined effect on the market return.

2. Are they adequately strong?

Yes, they are exceptionally strong.

The standard rule of thumb is that an F-statistic on the excluded instruments should be greater than 10. A value of 951.26 guarantees we do not have a weak instrument problem here.

3. What is the R^2 of the first-stage regression?

The Multiple R^2 is 0.9149.

This means that two instrumental variables, the ordinal rank of the returns (*RANK*) and the binary indicator for a positive return (*POS*), jointly explain approximately 91.5% of the variation in the market's excess return.

- (f) Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

The Hausman Test (Two Instruments)

- the p-value: 0.0287.
- The 1% Significance Level: $\alpha = 0.01$
- Conclusion: Because $0.0287 > 0.01$, we fail to reject the null hypothesis at the 1% level.

Can we conclude that the market return is exogenous?

Yes, strictly at the 1% level of significance, we conclude that the market return is exogenous. There is not enough statistical evidence at this strict threshold to prove that *mkt_ex* is correlated with the error term.

- (g) Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

1. The IV/2SLS Estimate

Looking at the output, the new estimated coefficient for *mkt_ex* using both *RANK* and *POS* as an instrumental variable is 1.283118.

2. Comparing IV to OLS

- OLS Estimate (Part a): 1.2018
- IV Estimate (Part g): 1.2831

The IV estimate is noticeably larger than the OLS estimate.

3. Does this agree with expectations?

Yes, it strongly agrees with our expectations.

Just like we saw in part (d), the presence of measurement error in the market return caused attenuation bias in the original OLS estimation, biasing the coefficient toward zero.

By using two very strong instruments to isolate the variation that is free from measurement error, we have corrected this downward bias. The true sensitivity (beta) of Microsoft to the market is actually higher (1.2831) than OLS initially suggested (1.2018).

- (h) Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

The manual Sargan test steps:

1. Extract the residuals from the 2SLS model in part (g).
2. Regress these residuals on all exogenous variables (which, in this simple CAPM case, are just two instruments: *RANK* and *POS*).
3. Calculate the test statistic: $n \times R^2$ (number of observations multiplied by the R-squared of this auxiliary regression).
4. Evaluate this statistic against a Chi-Square distribution with 1 degree of freedom (number of instruments minus number of endogenous variables).

1. The Test Results

Based on the manual regression of the 2SLS residuals on the two instruments,

- Sargan Test Statistic ($n \times R^2$): 0.5584634
- p-value: 0.45488

2. The Hypotheses

- Null Hypothesis (H_0): All instruments are valid. (Specifically, the surplus instrument is uncorrelated with the structural error term).
- Alternative Hypothesis (H_1): At least one instrument is invalid (correlated with the error term).

3. Conclusion

Because the p-value (0.45488) is significantly greater than 0.05, we fail to reject the null hypothesis at the 5% level of significance ($\alpha = 0.05$). We can confidently conclude that your instruments are valid. There is no statistical evidence to suggest that the surplus instrument is correlated with the error term. This means the choice to use both the ordinal rank (*RANK*) and the positive return indicator (*POS*) as instrumental variables was econometrically sound, and the 2SLS estimates from part (g) are

consistent and reliable.